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Task 1:

```
root@LAPTOP-0PA61QEN:~# ls /home
alurusaranya
root@LAPTOP-0PA61QEN:~# su alurusaranya
alurusaranya@LAPTOP-0PA61QEN:/root$ ls
ls: cannot open directory '.': Permission denied
alurusaranya@LAPTOP-0PA61QEN:/root$ cd ~
alurusaranya@LAPTOP-0PA61QEN:~$ pwd
/home/alurusaranya
alurusaranya@LAPTOP-0PA61QEN:~$ nano fork_exec.c
alurusaranya@LAPTOP-0PA61QEN:~$ gcc fork_exec.c -o fork_exec
alurusaranya@LAPTOP-0PA61QEN:~$ ./fork_exec
Enter a Linux command: ls

Parent Process
Parent PID : 462
Child PID  : 463

Child Process
Child PID : 463
Parent PID: 462

Executing command: ls
OSSP  fork_exec  fork_exec.c  hello  hello.c

Child process completed.
alurusaranya@LAPTOP-0PA61QEN:~$
```

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```
  GNU nano 7.2                                                        fork_exec.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    char command[100];
    pid_t pid;

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }

    if (pid == 0) {
        // Child process
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        printf("\nExecuting command: %s\n", command);

        execlp(command, command, NULL);

        // Executes only if execlp() fails
        perror("exec failed");
        exit(1);
```

```
    else {
        // Parent process
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID  : %d\n", pid);

        wait(NULL);

        printf("\nChild process completed.\n");
    }

    return 0;
}
```

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Task 2:

```
root@LAPTOP-0PA61QEN:~# ls /home
alurusaranya
root@LAPTOP-0PA61QEN:~# su alurusaranya
alurusaranya@LAPTOP-0PA61QEN:/root$ ls
ls: cannot open directory '.': Permission denied
alurusaranya@LAPTOP-0PA61QEN:/root$ cd ~
alurusaranya@LAPTOP-0PA61QEN:~$ pwd
/home/alurusaranya
alurusaranya@LAPTOP-0PA61QEN:~$ uname -a
Linux LAPTOP-0PA61QEN 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/
Linux
```

```
alurusaranya@LAPTOP-0PA61QEN:~$ lscpu
Architecture:             x86_64
  CPU op-mode(s):         32-bit, 64-bit
  Address sizes:          46 bits physical, 48 bits virtual
  Byte Order:             Little Endian
CPU(s):                   6
  On-line CPU(s) list:    0-5
Vendor ID:                GenuineIntel
  Model name:             Intel(R) Core(TM) Ultra 5 322
    CPU family:           6
    Model:                204
    Thread(s) per core:   1
    Core(s) per socket:   6
    Socket(s):            1
    Stepping:             3
    BogoMIPS:             6988.80
    Flags:                fpu vme de pse tsc msr pae mce cx8
                          apic sep mtrr pge mca cmov pat pse3
                          6 clflush mmx fxsr sse sse2 ss ht s
                          yscall nx pdpe1gb rdtscp lm constan
                          t_tsc rep_good nopl xtopology tsc_r
                          eliable nonstop_tsc cpuid tsc_known
                          _freq pni pclmulqdq vmx ssse3 fma c
                          x16 pcid sse4_1 sse4_2 x2apic movbe
                           popcnt tsc_deadline_timer aes xsav
                          e avx f16c rdrand hypervisor lahf_l
                          m abm 3dnowprefetch ssbd ibrs ibpb
                          stibp ibrs_enhanced tpr_shadow ept
                          vpid ept_ad fsgsbase tsc_adjust bmi
                          1 avx2 smep bmi2 erms invpcid rdsee
```

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```
Virtualization features:
  Virtualization:             VT-x
  Hypervisor vendor:          Microsoft
  Virtualization type:        full
Caches (sum of all):
  L1d:                        288 KiB (6 instances)
  L1i:                        384 KiB (6 instances)
  L2:                         18 MiB (6 instances)
  L3:                         12 MiB (1 instance)
NUMA:
  NUMA node(s):               1
  NUMA node0 CPU(s):          0-5
Vulnerabilities:
  Gather data sampling:       Not affected
  Ghostwrite:                 Not affected
  Indirect target selection:  Not affected
  Itlb multihit:              Not affected
  L1tf:                       Not affected
  Mds:                        Not affected
  Meltdown:                   Not affected
  Mmio stale data:            Not affected
  Old microcode:              Not affected
  Reg file data sampling:     Not affected
  Retbleed:                   Mitigation; Enhanced IBRS
  Spec rstack overflow:       Not affected
  Spec store bypass:          Mitigation; Speculative Store Bypas
                              s disabled via prctl
  Spectre v1:                 Mitigation; usercopy/swapgs barrier
                              s and __user pointer sanitization
  Spectre v2:                 Mitigation; Enhanced / Automatic IB
                              RS; IBPB conditional; PBRSB-eIBRS S
                              W sequence; BHI BHI_DIS_S
  Srbds:                      Not affected
  Tsa:                        Not affected
  Tsx async abort:            Not affected
  Vmscape:                    Not affected
```

```
alurusaranya@LAPTOP-0PA61QEN:~$ ps
    PID TTY          TIME CMD
    540 pts/0    00:00:00 bash
    560 pts/0    00:00:00 ps
```

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```
    560 pts/0    00:00:00 ps
alurusaranya@LAPTOP-0PA61QEN:~$ top
top - 05:59:52 up 6 min,  2 users,  load average: 0.00, 0.04, 0
Tasks:  26 total,   1 running,  25 sleeping,   0 stopped,   0 z
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,
MiB Mem :   7712.1 total,   6502.2 free,    491.1 used,    868.
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   7221.

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM
      1 root      20   0   21652  13072   9732 S   0.0   0.2
      2 root      20   0    3180   2200   2072 S   0.0   0.0
      6 root      20   0    3196   2128   2008 S   0.0   0.0
     52 root      19  -1   50360  16360  15284 S   0.0   0.2
    105 root      20   0   25604   6964   5128 S   0.0   0.1
    111 systemd+  20   0   21460  13272  11016 S   0.0   0.2
    112 systemd+  20   0   91028   7968   6996 S   0.0   0.1
    164 root      20   0    4244   2684   2432 S   0.0   0.0
    165 message+  20   0    9592   5328   4624 S   0.0   0.1
    172 root      20   0   18148   8920   7848 S   0.0   0.1
    175 root      20   0 1756104  13076  10888 S   0.0   0.2
    193 syslog    20   0  222516   5968   4556 S   0.0   0.1
    199 root      20   0    3124   1976   1836 S   0.0   0.0
    232 root      20   0  107016  23412  13948 S   0.0   0.3
    288 root      20   0    6672   4496   3728 S   0.0   0.1
    333 root      20   0   20312  11576   9452 S   0.0   0.1
    334 root      20   0   21164   3512   1796 S   0.0   0.0
    355 root      20   0    6080   5324   3696 S   0.0   0.1
    476 root      20   0    3188   1112    980 S   0.0   0.0
    477 root      20   0    3204   1252   1108 S   0.0   0.0
    481 root      20   0    6080   5336   3664 S   0.0   0.1
    514 root      20   0    6808   4444   3980 S   0.0   0.1
    517 alurusa+  20   0   20316  11544   9412 S   0.0   0.1
    518 alurusa+  20   0   21160   3580   1852 S   0.0   0.0
    540 alurusa+  20   0    6084   5300   3664 S   0.0   0.1
    561 alurusa+  20   0    9292   5412   3240 R   0.0   0.1
```